

AI-Powered Enhanced EHR Imaging & Documentation System

MILESTONE : 3

TEAM NAME : TEAM - B



Agenda



Structure Patient Data

Prepare and organize patient input for AI processing.



Generate Clinical Notes

Utilize Generative AI for automated note creation.



Automate ICD-10 Mapping

Efficiently map diagnosis codes for billing and records.



Validate Output

Ensure accuracy and consistency of all generated data.

The Problem Statement

Manual clinical documentation and ICD-10 code assignment consume significant time and are prone to human errors. Doctors must write lengthy notes and manually search through large ICD-10 databases, leading to :

Inconsistent Documentation

Lack of standardization across patient records.

Delayed Diagnosis Workflows

Slower access to critical patient information.

Reduced Clinical Efficiency

Healthcare professionals spending less time on patient care.

Poorly Structured EHR Data

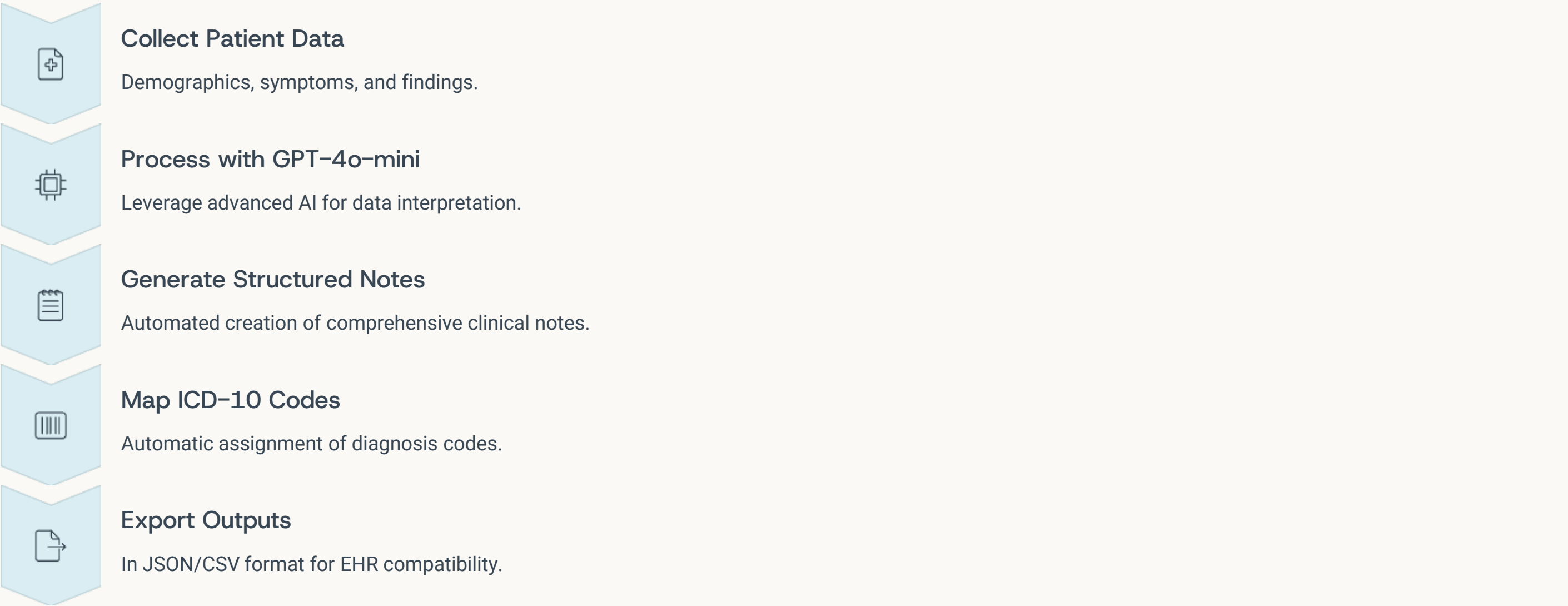
Challenges in data analysis and interoperability.

There is a strong need for an automated, intelligent, and structured documentation system to overcome these challenges.

Proposed Solution: AI-Powered Automation

Our system utilizes Generative AI to automatically create clinical notes and map ICD-10 diagnosis codes from structured patient input. This approach ensures accuracy, speed, and standardization for seamless EHR integration.

Workflow Overview





Tools & Technologies Utilized

Python

Core programming language for system development.

Azure OpenAI

GPT-4o-mini / GPT-4 for clinical note generation.

Pandas / JSON / CSV

For efficient structured data handling.

VS Code / Google Colab

Primary environments for development and testing.

Implementation Progress & Achievements

1. Patient Input Formatting

Process for structured data input is fully completed.

2. GPT-4o – mini Integration

Seamless integration for medical text generation.

3. Automated Clinical Notes

System successfully generates complete clinical notes.

4. ICD-10 Mapping Module

Robust implementation for accurate code mapping.

5. Output File Storage

Generated outputs are stored in JSON/CSV formats.

6. Evaluation Phase

Initiated to validate system performance.

Technical Challenges & Solutions

1. Medical Terminology Complexity

The intricate nature of medical language posed a challenge for AI interpretation.

Solution: Enhanced Prompt Engineering

Implemented stronger prompt engineering and sample-based prompting for improved contextual understanding.

2. Incorrect ICD-10 Mapping

Challenges arose in cases with similar symptoms leading to potential mismapping.

Solution: Validation & Post-processing

Added comprehensive validation checks and post-processing logic to refine mapping accuracy.



Key Learnings & Achieved Results

Improved Data Understanding

Deeper insight into structured healthcare datasets.

Enhanced AI Generation

Better clinical note generation through refined prompt engineering.

Reliable ICD-10 Mapping

Increased accuracy and consistency of coding outputs.

System Outcomes

- Successfully generates detailed clinical notes.
- Accurate ICD-10 code predictions on sample patient data.
- Outputs exported in structured, EMR-ready formats.
- Significant reduction in time compared to manual documentation.

Next Steps

Future Directions

- Build a simple user interface for testing the workflow, enhancing user interaction.
- Begin Milestone 4: Integration with previous modules to create a cohesive system.

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